

ECA5612 –Advanced Wireless Communication Systems 5G / 6G Communication Systems.

Ex.No. 11: Implementation of IoT-based Smart Agriculture using MATLAB.

Objective 1 Matlab Code and Output:

Program:

```
clc; clear; close all;

% Simulated soil moisture data (%)

time = 1:10;

soil_moisture = [30 35 28 25 20 18 22 30 40 45];

% Threshold for irrigation

threshold = 40;

figure

% ----- Graph 1: Soil Moisture Variation -----

subplot(2,1,1)

plot(time, soil_moisture, '-o', 'LineWidth', 2)

title('Soil Moisture Monitoring in IoT Smart Agriculture')

xlabel('Time (Hours)')

ylabel('Soil Moisture (%)')

grid on

hold on

yline(threshold,'r--','Threshold')

% ----- Graph 2: Moisture Status (Dry/Wet Level) -----

status = soil_moisture >= threshold;

subplot(2,1,2)

stem(time, status, 'filled', 'LineWidth', 2)

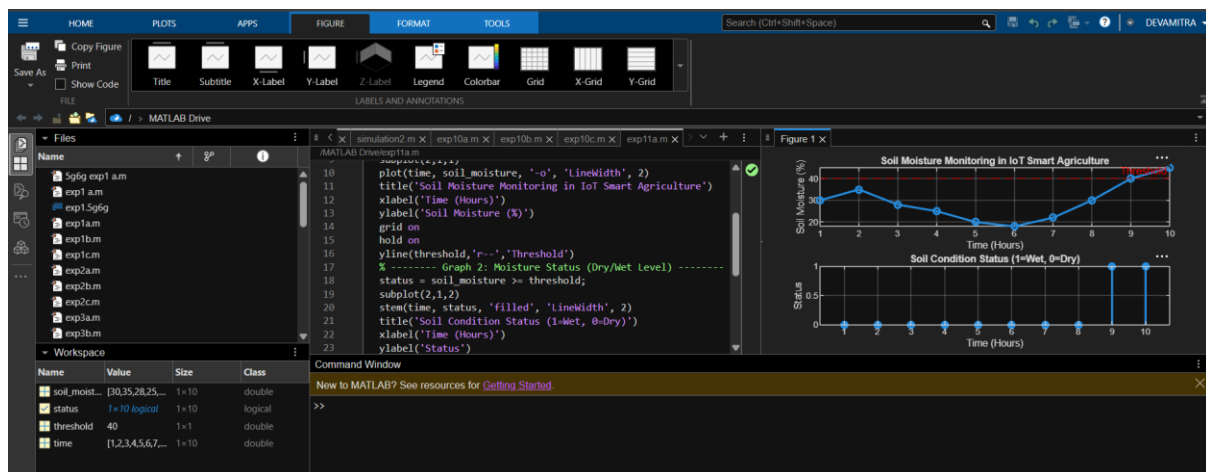
title('Soil Condition Status (1=Wet, 0=Dry)')

xlabel('Time (Hours)')

ylabel('Status')

grid on
```

Output:



Objective 2 Matlab Code and Output:

Program:

```
clc; clear; close all;
```

```
% Simulated soil moisture (%)
```

```
time = 1:10;
```

```
soil_moisture = [30 35 28 25 20 18 22 30 40 45];
```

```
% Threshold level
```

```
threshold = 40;
```

```
% Pump control logic (1 = ON, 0 = OFF)
```

```
pump_status = soil_moisture < threshold;
```

```
figure
```

```
% ----- Graph 1: Soil Moisture vs Pump Status -----
```

```
subplot(2,1,1)
```

```
plot(time, soil_moisture, '-o', 'LineWidth', 2)
```

```
title('Soil Moisture vs Pump Control')
```

```
xlabel('Time (Hours)')
```

```
ylabel('Soil Moisture (%)')
```

```
grid on
```

```
hold on
```

```

yline(threshold,'r--','Threshold')

% ----- Graph 2: Pump Status (ON/OFF) -----

subplot(2,1,2)

stem(time, pump_status, 'filled','LineWidth',2)

title('Water Pump Status (1=ON, 0=OFF)')

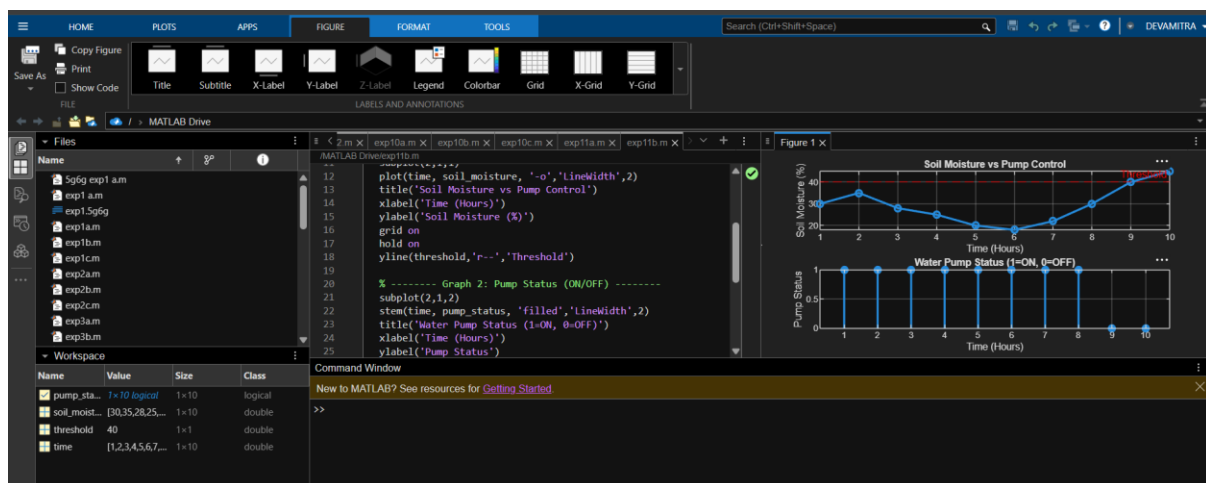
xlabel('Time (Hours)')

ylabel('Pump Status')

grid on

```

Output:



Objective 3 Matlab Code and Output:

Program:

```

clc; clear; close all;

% Soil moisture data

time = 1:10;

soil = [30 35 28 25 20 18 22 30 40 45];

threshold = 40;

% Irrigation efficiency

efficiency = min((soil / threshold) * 100, 100);

figure;

```

```

% Soil moisture graph

subplot(2,1,1);

plot(time, soil, '-o', 'LineWidth', 2);

hold on;

yline(threshold, 'r--', 'Threshold');

title('Soil Moisture Variation');

xlabel('Time (Hours)');

ylabel('Soil Moisture (%)');

grid on;

% Efficiency graph

subplot(2,1,2);

plot(time, efficiency, '-s', 'LineWidth', 2);

title('Irrigation Efficiency');

xlabel('Time (Hours)');

ylabel('Efficiency (%)');

grid on;

```

Output:

